

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES				
10							*****VALVES*****			810	NIPPLE TO BE CUT FROM SCH 160 PIPE.			
20										816	COMPACT FLANGES CAN BE USED FOR TIE-IN PURPOSE ONLY TO EXISTING PIPING AND EQUIPMENT.			
20							***** MODULAR VALVES *****							
30														
30			MODULAR DBB	K06AM10	0.5	1.5	MODULAR VALVE DB&B BALL TYPE 600# RF INC 825 INC 625 TRIM METAL LO - FULL SIZE TYPE 2BV	0	0	817	USE ONLY FOR TIE IN			
40			MODULAR DBB	K06AM14	2	2	MODULAR VALVE DB&B BALL TYPE 600# RF INC 825 INC 625 TRIM METAL LO - FULL SIZE TYPE 2BV	0	0	B1	ALL BOLTS ARE METRIC DIA.			
50		VAV	MODULAR DBB	K06AM14	3	3	MODULAR VALVE DB&B BALL TYPE 600# RF INC 825 INC 625 TRIM METAL LO - FULL SIZE TYPE 2BV	0	0	B2	7/8" X 145MM BOLTS ONLY TO BE USED FOR 1			
60			MODULAR DBB	M06AM10	0.75	1.5	MODULAR VALVE DB&B BALL TYPE 600# RF INC 825 INC 625 TRIM METAL LO - CONTROLLED BORE TYPE 2BVR	0	0		INCH 1500# PERAR BALL VALVES ITEM CODE B15AM10			
70			MODULAR DBB	M06AM14	2	2	MODULAR VALVE DB&B BALL TYPE 600# RF INC 825 INC 625 TRIM METAL LO - CONTROLLED BORE TYPE 2BVR	0	0	B5	7/8" X 155MM BOLTS ONLY TO BE USED FOR 1			
80		VAW	MODULAR DBB	M06EM10	0.75	1	MODULAR VALVE DB&B BALL TYPE 600# RF INC 825 INC TRIM 625 METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR	2	2		INCH 1500# PERAR BALL VALVES ITEM CODE B15AC02 WHEREVER MATCHED WITH A SPECTACLE BLIND			
90		VAW	MODULAR DBB	M06EM10	1	1	MODULAR VALVE DB&B BALL TYPE 600# RF INC 825 INC TRIM 625 METAL LO REDUCING - CONTROLLED BORE TYPE 2BVR	3	3	B6	7/8" X 145MM BOLTS ONLY TO BE USED FOR 1			
100											INCH 1500# PERAR BALL VALVES ITEM CODE B15AS22			
100		VB2					***** BALL VALVES *****			FA8	USE THREADED FLANGES FOR INSTRUMENT CONNECTIONS ONLY.			
110														
110			BALL	A06AM13	2	3	BALL 600# RF 825 INCONEL 625 TRIM TRUNNION SOFT RB LO	0	0	FF5	FLAT BOTTOM NIPOFLANGES FOR USE ON WELD CAPS AND BLIND FLANGES			
120			BALL	A06AR13	4	4	BALL 600# RF LTCS + CLAD INC 825 INC 625 TRIM TRUNNION SOFT RB LO	0	0					
130			BALL	A06AR15	6	16	BALL 600# RF LTCS + CLAD 825 INCONEL 625 TRIM TRUNNION SOFT RB GO	0	0	ND9	USE ONLY WITH CONSENT OF MATERIALS ENGINEER.			
140		VA5	BALL	B06AM10	1	1	BALL 600# RF 825 INCONEL 625 TRIM FLOATING METAL FB LO	0	0					
150			BALL	B06AM09	0.75	1.5	BALL 600# RF 825 INCONEL 625 TRIM FLOATING SOFT FB LO	0	0	PS1	PIPE ONLY TO BE USED AS TRUNNION			
160		VA5	BALL	B06AM13	2	4	BALL 600# RF 825 INCONEL 625 TRIM TRUNNION SOFT FB LO	0	0		SUPPORTS			
170	U04	VA5	BALL	B06AR15	4	16	BALL 600# RF LTCS + CLAD 825 INCONEL 625 TRIM TRUNNION SOFT FB GO	0	0	PS4	PIPE ONLY TO BE USED AS TRUNNION			
180	U04	VA5	BALL	A06AM14	2	2	BALL 600# RF 825 INCONEL 625 TRIM TRUNNION METAL RB LO	0	0		SUPPORTS. USE FOR HEADER SIZE 4"AND ABOVE ONLY.			
190							***** RISING STEM BALL VALVES *****			SB5	USE JACK SCREWS ONLY WHERE PIPING CAN NOT BE OPENED WITH FLANGE SPREADER.			
190										SB6	USE JACKSCREWS FOR PADDLE			
200		VA5	BALL	R06AL16	8	16	BALL RISING STEM 600# RF 625 INCONEL 625 TRIM TRUNNION METAL RB GO	0	0		BLANKS/SPACERS AND FLANGE SPREADERS FOR SPECTACLE BLINDS.			
210	U04	VA5	BALL	R06AM14	3	3	BALL RISING STEM 600# RF 825 INCONEL 625 TRIM TRUNNION METAL RB HO	0	0	UJ2	USE 1 1/2IN WITH THERMOWELL			
220	U04	VA5	BALL	R06AN16	8	10	BALL RISING STEM 600# RF LTCS + CLAD INC 625 TRIM 625 TRUNNION METAL RB GO	0	0		CONNECTIONS ONLY & 3/4IN (IF SHOWN) FOR DBB PRESSURE CONNECTIONS WITH CHEMICAL SEALS.			
230	U04	VAZ	BALL	R06AN16	12	16	BALL RISING STEM 600# RF LTCS + CLAD INC 625 TRIM 625 TRUNNION METAL RB GO	0	0	VA5	USE THESE VALVES ONLY WHERE SPECIFIED			
240							***** GLOBE VALVES *****				ON PIPING AND INSTRUMENT DIAGRAMS			
240										VAV	USAGE IS LIMITED TO INSTRUMENT ISOLATION PURPOSE.			
250			GLOBE	L06AM43	0.75	3	GLOBE 600# RF INC 825 INC TRIM 625 OS&Y HW	0	0	VAW	VALVE ONLY TO BE USED IN ACCORDANCE WITH DOCUMENT L-ST-2055 - VENT DRAIN AND INSTRUMENT CONNECTION DETAILS.			
260			GLOBE	L06AR43	4	4	GLOBE 600# RF LTCS A352 LCC + 825 CLAD INC 625 TRIM OS&Y HW	0	0					
270			GLOBE	L06AR44	6	8	GLOBE 600# RF LTCS A352 LCC + 825 CLAD INC 625 TRIM OS&Y GO	0	0	VAZ	USE THESE VALVES ONLY WHERE CALLED FOR ON FLOW DIAGRAMS AND/OR DRAWINGS.			
280							***** CHECK VALVES *****				THESE VALVES MUST BE INSTALLED WITH THE STEM IN VERTICAL POSITION ONLY.			
280										VB2	SOFT SEATED BALL VALVES ARE LIMITED TO MAXIMUM TEMPERATURE OF 200°C.			
290										VC1	SPRING PISTON CHECK VALVES ARE PREFERRED IN THE HORIZONTAL POSITION, BUT MAY BE USED IN THE VERTICAL POSITION WITH ENGINEERING APPROVAL.			
290		VC1	CHECK	C06AM51	0.75	1.5	CHECK PISTON 600# RF INC 825 TRIM 625	0	0					
300	U04		CHECK	C06AM46	2	3	CHECK DUAL WAFER LUG 600# RF INCONEL 825 N08825 THROU BOLTED	0	0					
310	U04		CHECK	C06AR46	4	8	CHECK DUAL WAFER LUG 600# RF LTCS + CLAD INC 825 TRIM 625 THROU BOLTED	0	0					
320	U04		CHECK	C06AR53	10	16	CHECK DUAL DBL FLG 600# RF LTCS A352 LCC + 825 CLAD TRIM 625	0	0					
330							*****PIPES*****							
330														
340			PIPE	5732001	0.75	1	PIPE SCH 80S SMLS ALLOY 825 B423 UNS N08825 PE	0	0					
350			PIPE	5732002	1.5	1.5	PIPE SCH 40S SMLS ALLOY 825 B423 UNS N08825 PE	0	0	VC1	SPRING PISTON CHECK VALVES ARE PREFERRED IN THE HORIZONTAL POSITION, BUT MAY BE USED IN THE VERTICAL POSITION WITH ENGINEERING APPROVAL.			
360			PIPE	5361603	2	3	PIPE SCH 40S SMLS ALLOY 825 B423 UNS N08825 BE	0	0					
370			PIPE	5358514_C80	4	8	PIPE SCH XS SMLS LTCS A333 GR.6 WITH 3 MM INC 825 INTERNAL CLADDING	0	0					
380			PIPE	5358507_C80	10	16	PIPE SCH 80 SMLS LTCS A333 GR.6 WITH 3MM INC 825 INTERNAL CLADDING	0	0					
390		810	PIPE	5732003	0.5	0.5	PIPE SCH 160 SMLS ALLOY 825 B423 UNS N08825 PE	0	0					
400		810	PIPE	5732003	0.75	0.75	PIPE SCH 160 SMLS ALLOY 825 B423 UNS N08825 PE	0	0					
410	U04		PIPE	5358510_C60	4	4	PIPE SCH 120 SMLS LTCS A333 GR.6 BE WITH 3MM INC 625 INTERNAL CLADDING	0	0					
420	U04		PIPE	5358599_C60	18	24	PIPE SCH 80 EFW LTCS A671 CC60 CL.22 BE WITH 3MM INC 825/625 INTERNAL CLADDING	0	0					
430	U04		PIPE	5358507_C60	10	16	PIPE SCH 80 SMLS LTCS A333 GR.6 WITH 3MM INC 625 INTERNAL CLADDING	0	0					
440	U04		PIPE	5358514_C60	4	8	PIPE SCH XS SMLS LTCS A333 GR.6 WITH 3 MM INC 625 INTERNAL CLADDING	0	0					
450	U04		PIPE	5358623_C60	30	30	PIPE WT 35.0MM EFW LTCS A671 CC60 CL.22 BE WITH 3 MM INC 825/625 INTERNAL CLADDING	0	0					
460							***** TRUNNION PIPE SUPPORTS *****							
460														
470														
470	U04	PS1	PIPE	5358510	4	4	PIPE SCH 120 SMLS LTCS A333 GR.6 BE	0	0					
480	U04	PS4	PIPE	5358504	3	3	PIPE SCH 40 SMLS LTCS A333GR.6 BE	0	0					
490	U04	PS1	PIPE	5358507	10	16	PIPE SCH 80 SMLS LTCS A333 GR.6 BE	0	0					
500	U04	PS1	PIPE	5358514	6	8	PIPE SCH XS SMLS LTCS A333 GR.6 BE	0	0					
510							***** FITTINGS *****							
510														
520														
520			ELBOW	5425111	0.75	1	ELL 90 DEG LR SCH 80S ALLOY 825 B366-WPS UNS N08825	0	0					
530			ELBOW	5424722	1.5	3	ELL 90 DEG LR SCH 40S ALLOY 825 B366-WPS UNS N08825	0	0					
			CONSTRUCTION		CORR.ALLOW.		SERVICE	MATERIAL		DATE	BY	REV	SHEET	PIPING LINE CLASS
			ALL SIZES - BUTT WELD PWHT FOR SIZES 4" AND LARGER SEAL WELDING - FOR THREADED CONNECTIONS ONLY PIPE BENDS - NONE THREAD SEALANT - PTFE PASTE		CORR. ALLOW. 1 MM		SOUR SERVICE - NACE MR0175/ ISO 15156 DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.3 FLUID CODES - PIM-SU-5112-TCO	MATERIAL: LTCS + CRA FLANGE RATING: 600 RF TEMPERATURE RANGE: -40 / 200 °C		12/16/2024	BY OZAC	U04	Page 1 of 4	600C80

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES
540	U04		ELBOW	5392539_C60	30	30	ELL 90 DEG LR WT 35.0MM LTCS A420 WPL6-W WITH 3MM INC 625 INTERNAL CLADDING	0	0	
550			ELBOW	5392600_C80	4	8	ELL BW 90 DEG LR SCH XS LTCS A420 WPL6 WITH 3 MM INC 825 INTERNAL CLADDING	0	0	
560			ELBOW	5392546_C80	10	16	ELL BW 90 DEG LR SCH 80 LTCS A420 WPL6 WITH 3 MM INC 825 INTERNAL CLADDING	0	0	
570	U04		ELBOW	5392700_C60	4	4	ELL BW 90 DEG LR SCH 120 LTCS A420 WPL6 WITH 3MM INC 625 INTERNAL CLADDING	0	0	
580	U04		ELBOW	5392546_C60	10	16	ELL BW 90 DEG LR SCH 80 LTCS A420 WPL6 WITH 3 MM INC 625 INTERNAL CLADDING	0	0	
590	U04		ELBOW	5392600_C60	4	8	ELL BW 90 DEG LR SCH XS LTCS A420 WPL6 WITH 3 MM INC 625 INTERNAL CLADDING	0	0	
600			ELBOW	5425113	0.75	1	ELL 45 DEG SCH 80S ALLOY 825 B366-WPS UNS N08825	0	0	
610			ELBOW	5424727	1.5	3	ELL 45 DEG SCH 40S ALLOY 825 B366-WPS UNS N08825	0	0	
620			ELBOW	5392606_C80	4	8	ELL BW 45 DEG LR SCH XS LTCS A420 WPL6 WITH 3 MM INC 825 INTERNAL CLADDING	0	0	
630			ELBOW	5392552_C80	10	16	ELL BW 45 DEG LR SCH 80 LTCS A420 WPL6 WITH 3 MM INC 825 INTERNAL CLADDING	0	0	
640	U04		ELBOW	5392701_C60	4	4	ELL BW 45 DEG LR SCH 120 LTCS A420 WPL6 WITH 3MM INC 625 INTERNAL CLADDING	0	0	
650	U04		ELBOW	5392552_C60	10	16	ELL BW 45 DEG LR SCH 80 LTCS A420 WPL6 WITH 3 MM INC 625 INTERNAL CLADDING	0	0	
660	U04		ELBOW	5392606_C60	4	8	ELL BW 45 DEG LR SCH XS LTCS A420 WPL6 WITH 3 MM INC 625 INTERNAL CLADDING	0	0	
670										
670			TEE	5425119	0.75	1	TEE SCH 80S ALLOY 825 B366-WPS UNS N08825	0	0	
680			TEE	5424725	1.5	3	TEE SCH 40S ALLOY 825 B366-WPS UNS N08825	0	0	
690			TEE	5392609_C80	4	8	TEE BW SCH XS LTCS A420 WPL6 WITH 3 MM INC 825 INTERNAL CLADDING	0	0	
700			TEE	5392555_C80	10	16	TEE BW SCH 80 LTCS A420 WPL6 WITH 3 MM INC 825 INTERNAL CLADDING	0	0	
710	U04		TEE	5392702_C60	4	4	TEE BW SCH 120 LTCS A420 WPL6 WITH 3MM INC 625 INTERNAL CLADDING	0	0	
720	U04		TEE	5392555_C60	10	16	TEE BW SCH 80 LTCS A420 WPL6 WITH 3 MM INC 625 INTERNAL CLADDING	0	0	
730	U04		TEE	5392609_C60	4	8	TEE BW SCH XS LTCS A420 WPL6 WITH 3 MM INC 625 INTERNAL CLADDING	0	0	
740			TEE RED	5425126	0.75	0.75	TEE RED SCH 80S ALLOY 825 B366-WPS UNS N08825	1	1	
750			TEE RED	5425124	0.75	1	TEE RED SCH 40S/80S ALLOY 825 B366-WPS UNS N08825	1.5	2	
760			TEE RED	5424724	1.5	2	TEE RED SCH 40S ALLOY 825 B366-WPS UNS N08825	2	3	
770			TEE RED	5392413_C80	4	6	TEE RED BW SCH XS LTCS A420 WPL6 WITH 3 MM INC 825 INTERNAL CLADDING	6	8	
780			TEE RED	5392463_C80	4	8	TEE RED BW SCH 80/XS LTCS A420 WPL6 WITH 3 MM INC 825 INTERNAL CLADDING	10	16	
790			TEE RED	5392464_C80	10	14	TEE RED BW SCH 80 LTCS A420 WPL6 WITH 3 MM INC 825 INTERNAL CLADDING	12	16	
800	U04		TEE RED	5392464_C60	10	14	TEE RED BW SCH 80 LTCS A420 WPL6 WITH 3 MM INC 625 INTERNAL CLADDING	12	16	
810	U04		TEE RED	5392577_C60	4	4	TEE RED BW SCH 80/120 LTCS A420 WPL6 WITH 3 MM INC 625 INTERNAL CLADDING	10	10	
820	U04		TEE RED	5392463_C60	4	8	TEE RED BW SCH 80/XS LTCS A420 WPL6 WITH 3 MM INC 625 INTERNAL CLADDING	10	16	
830	U04		TEE RED	5392413_C60	4	6	TEE RED BW SCH XS LTCS A420 WPL6 WITH 3 MM INC 625 INTERNAL CLADDING	6	8	
840	U04		TEE RED	5392626_C60	4	4	TEE RED BW SCH XS/SCH 120 LTCS A420 WPL6 WITH 3 MM INC 625 INTERNAL CLADDING	6	8	
850										
850			CAP	5424270	0.75	1	CAP SCH 80S ALLOY 825 B366 UNS N08825	0	0	
860			CAP	5424219	1.5	3	CAP SCH 40S ALLOY 825 B366 UNS N08825	0	0	
870			CAP	5392610_C80	4	8	CAP BW SCH XS LTCS A420 WPL6 WITH 3 MM INC 825 INTERNAL CLADDING	0	0	
880			CAP	5392556_C80	10	16	CAP BW SCH 80 LTCS STL A420 WPL6 WITH 3 MM INC 825 INTERNAL CLADDING	0	0	
890	U04		CAP	5392556_C60	10	16	CAP BW SCH 80 LTCS STL A420 WPL6 WITH 3 MM INC 625 INTERNAL CLADDING	0	0	
900	U04		CAP	5392610_C60	4	8	CAP BW SCH XS LTCS A420 WPL6 WITH 3 MM INC 625 INTERNAL CLADDING	0	0	
910	U04		CAP	5392703_C60	4	4	CAP SCH 120 LTCS A420 WPL6 WITH 3MM INC 625 INTERNAL CLADDING	0	0	
920										
920			RED CONC	5425122	0.75	0.75	REDUCER CONC SCH 80S ALLOY 825 B366-WPS N08825	1	1	
930			RED CONC	5425143	0.75	1	REDUCER CONC SCH 40S/80S ALLOY 825 B366-WPS N08825	1.5	2	
940			RED CONC	5424733	1.5	2	REDUCER CONC SCH 40S ALLOY 825 B366-WPS N08825	2	3	
950			RED CONC	5425149	1.5	3	REDUCER CONC SCH XS/40S ALLOY 825 B366-WPS N08825	4	6	
960			RED CONC	5392458_C80	4	6	REDUCER CONC SCH XS LTCS A420 WPL6 WITH 3 MM INC 825 INTERNAL CLADDING	6	8	
970			RED CONC	5392309_C80	8	8	REDUCER CONC SCH 80/XS LTCS A420 WPL6 WITH 3 MM INC 825 INTERNAL CLADDING	10	16	
980			RED CONC	5392559_C80	10	14	REDUCER CONC SCH 80 LTCS A420 WPL6-W WITH 3 MM INC 825 INTERNAL CLADDING	12	16	
990	U04		RED CONC	5425161	3	3	REDUCER CONC SCH 120/40S ALLOY 825 B366-WPS N08825	6	6	
1000	U04		RED CONC	5392741	1.5	3	REDUCER CONC SCH 160/40S ALLOY 825 B366-WPS N08825	4	4	
1010	U04		RED CONC	5392559_C60	10	14	REDUCER CONC SCH 80 LTCS A420 WPL6-W WITH 3 MM INC 625 INTERNAL CLADDING	12	16	
1020	U04		RED CONC	5392309_C60	8	8	REDUCER CONC SCH 80/XS LTCS A420 WPL6 WITH 3 MM INC 625 INTERNAL CLADDING	10	16	
1030	U04		RED CONC	5392458_C60	4	6	REDUCER CONC SCH XS LTCS A420 WPL6 WITH 3 MM INC 625 INTERNAL CLADDING	6	8	
1040	U04		RED CONC	5392740_C60	4	4	REDUCER CONC SCH XS/120 LTCS A420 WPL6 WITH 3 MM INC 625 INTERNAL CLADDING	6	8	
1050	U04		RED CONC	5392369_C60	20	20	REDUCER CONC WT 35.0MM/SCH 80 LTCS A420 WPL6-W WITH 3 MM INC 625 INTERNAL CLADDING	30	30	
1060			RED ECC	5425123	0.75	0.75	REDUCER ECC SCH 80S ALLOY 825 B366-WPS N08825	1	1	
1070			RED ECC	5425145	0.75	1	REDUCER ECC SCH 40S/80S ALLOY 825 B366-WPS N08825	1.5	2	
1080			RED ECC	5424734	1.5	2	REDUCER ECC SCH 40S ALLOY 825 B366-WPS N08825	2	3	
1090			RED ECC	5425160	1.5	3	REDUCER ECC SCH XS/40S ALLOY 825 B366-WPS N08825	4	6	
1100			RED ECC	5392461_C80	4	6	REDUCER ECC SCH XS LTCS A420 WPL6 WITH 3 MM INC 825 INTERNAL CLADDING	6	8	
1110			RED ECC	5392311_C80	8	8	REDUCER ECC SCH 80/XS LTCS A420 WPL6 WITH 3 MM INC 825 INTERNAL CLADDING	10	16	
1120			RED ECC	5392313_C80	10	14	REDUCER ECC SCH 80 LTCS A420 WPL6-W WITH 3 MM INC 825 INTERNAL CLADDING	12	16	
1130	U04		RED ECC	5425162	3	3	REDUCER ECC SCH 120/40S ALLOY 825 B366-WPS N08825	6	6	
1140	U04		RED ECC	5425159	1.5	3	REDUCER ECC SCH 160/40S ALLOY 825 B366-WPS N08825	4	4	
1150	U04		RED ECC	5392313_C60	10	14	REDUCER ECC SCH 80 LTCS A420 WPL6-W WITH 3 MM INC 625 INTERNAL CLADDING	12	16	
1160	U04		RED ECC	5392311_C60	8	8	REDUCER ECC SCH 80/XS LTCS A420 WPL6 WITH 3 MM INC 625 INTERNAL CLADDING	10	16	
1170	U04		RED ECC	5392461_C60	4	6	REDUCER ECC SCH XS LTCS A420 WPL6 WITH 3 MM INC 625 INTERNAL CLADDING	6	8	
1180	U04		RED ECC	5392820_C60	4	4	REDUCER ECC SCH XS/SCH 120 LTCS A420 WPL6 WITH 3 MM INC 625 INTERNAL CLADDING	6	8	
1190										
1190			OLETS	5513489	0.75	1	NIPOFLANGE 600# RF SCH 40S/80S ALLOY 825 B564 UNS N08825	2	3	
1200			OLETS	5513490	0.75	1	NIPOFLANGE 600# RF SCH XS/80S ALLOY 825 B564 UNS N08825	4	8	
1210			OLETS	5513480	1.5	2	NIPOFLANGE 600# RF SCH XS/40S ALLOY 825 B564 UNS N08825	4	8	
1220			OLETS	5513500	0.75	1	NIPOFLANGE 600# RF SCH 80/80S ALLOY 825 B564 UNS N08825	10	16	

	TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES - BUTT WELD PWHT FOR SIZES 4" AND LARGER SEAL WELDING - FOR THREADED CONNECTIONS ONLY PIPE BENDS - NONE THREAD SEALANT - PTFE PASTE	CORR. ALLOW. 1 MM	SOUR SERVICE - NACE MR0175/ ISO 15156 DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.3 FLUID CODES - PIM-SU-5112-TCO	MATERIAL: LTCS + CRA FLANGE RATING: 600 RF TEMPERATURE RANGE: -40 / 200 °C	12/16/2024	BY OZAC	U04	Page 2 of 4	600C80

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES
1230			OLETS	5513501	1.5	2	NIPOFLANGE 600# RF SCH 80/40S ALLOY 825 B564 UNS N08825	10	16	
1240		UJ2	OLETS	5513510	1.5	1.5	NIPOFLANGE 600# RF SCH XS/160 ALLOY 825 B564 UNS N08825	4	8	
1250		UJ2	OLETS	5513512	1.5	1.5	NIPOFLANGE 600# RF SCH 80/160 ALLOY 825 B564 UNS N08825	10	16	
1260	U04	UJ2	OLETS	5513405	1.5	1.5	NIPOFLANGE 600# RF SCH 120/160 ALLOY 825 B564 UNS N08825	4	4	
1270	U04		OLETS	5513404	1.5	2	NIPOFLANGE 600# RF SCH 120/40S ALLOY 825 B564 UNS N08825	4	4	
1280	U04		OLETS	5513402	0.75	1	NIPOFLANGE 600# RF SCH 120/80S ALLOY 825 B564 UNS N08825	4	4	
1290		FF5	OLETS	5516491	0.75	1	NIPOFLANGE FLAT BOTTOM 600# RF SCH 80S ALLOY 825 B564 UNS N08825	0	0	
1300										
1300			OLETS	5513413	0.75	1	WELDOLET SCH 40S/80S ALLOY 825 B564 UNS N08825	2	3	
1310			OLETS	5513415	0.75	1	WELDOLET SCH XS/80S ALLOY 825 B564 UNS N08825	4	8	
1320			OLETS	5513419	1.5	3	WELDOLET SCH XS/ 40S ALLOY 825 B564 UNS N08825	4	8	
1330			OLETS	5513445	0.75	1	WELDOLET SCH 80/80S ALLOY 825 B564 UNS N08825	10	16	
1340			OLETS	5513446	1.5	3	WELDOLET SCH 80/40S ALLOY 825 B564 UNS N08825	10	16	
1350	U04		OLETS	5505262_C80	4	6	WELDOLET SCH 80/XS LTCS A350 LF2 CL.1 WITH 3MM INC 825 INTERNAL CLADDING	10	16	
1360	U04		OLETS	5509610	1.5	3	WELDOLET SCH 120 /40S ALLOY 825 B564 UNS N08825	4	4	
1370	U04		OLETS	5509620	0.75	1	WELDOLET SCH 120 /80S ALLOY 825 B564 UNS N08825	4	4	
1380	U04		OLETS	5509619_C60	4	4	WELDOLET SCH 80/120 LTCS A350 LF2 CL.1 WITH 3MM INC 625 INTERNAL CLADDING	10	16	
1390	U04		OLETS	5513448	2	2	WELDOLET WT 35.0MM/SCH 40S ALLOY 825 B564 UNS N08825	30	30	
1400										
1400		UJ2	ELBOFLG	5516673	1.5	1.5	ELBOFLANGE 600# RF SCH XS/ SCH 160 ALLOY 825 B564 UNS N08825	4	8	
1410		UJ2	ELBOFLG	5516674	1.5	1.5	ELBOFLANGE 600# RF SCH 80/ SCH 160 ALLOY 825 B564 UNS N08825	10	16	
1420	U04	UJ2	ELBOFLG	5516676	1.5	1.5	ELBOFLANGE 600# RF SCH 120/ SCH 160 ALLOY 825 B564 UNS N08825	4	4	
1430										
1430							***** FLANGES *****			
1440										
1440		FA8	TEE RED	5614300	0.5	0.5	FLG SCRDR (FNPT) REDUCING 600# RF ALLOY 825 B564 UNS N08825	0.75	1.5	
1450										
1450			FLANGE	5851204	0.5	2	FLANGE ADAPTER 600# RF 316 SS X 12 MM TUBE	0	0	
1460	U04	ND9	FLG ADAPTER	5851214	0.5	2	FLANGE ADAPTER 600# RF ALLOY 825 B564 UNS N08825 X 12 MM TUBE	0	0	
1470										
1470			FLANGE	5569391	0.75	1	FLG WN 600# RF ALLOY 825 SCH 80S B564 UNS N08825	0	0	
1480			FLANGE	5569392	1.5	3	FLG WN 600# RF ALLOY 825 SCH 40S B564 UNS N08825	0	0	
1490	U04		FLANGE	5557036_C80	4	8	FLG WN 600# RF LTCS A350 LF2 CL.1 SCH XS WITH 3MM INC 825 CLAD FACE & BORE	0	0	
1500	U04		FLANGE	5557040_C80	10	16	FLG WN 600# RF LTCS A350 LF2 CL.1 SCH 80 WITH 3MM INC 825 CLAD FACE & BORE	0	0	
1510			BLIND	5617167	0.5	3	FLG BLIND 600# RF ALLOY 825 B564 UNS N08825	0	0	
1520	U04		BLIND	5615598_C80	4	16	FLG BLIND 600# RF LTCS A350 LF2 CL.1 WITH 3 MM INC 825 CLAD FACE	0	0	
1530	U04		BLIND	5615598_C60	4	16	FLG BLIND 600# RF LTCS A350 LF2 CL.1 WITH 3 MM INC 625 CLAD FACE	0	0	
1540	U04	817	FLANGE	5619802_C60	30	30	FLG COMPACT 600# RJ LTCS A350 LF2 CL.1 WT 35.0MM WITH 3MM INC 625 CLAD FACE & BORE LINE CLASS 600C80	0	0	
1550	U04		FLANGE	5557031_C60	4	4	FLG WN 600# RF LTCS A350 LF2 CL.1 SCH 120 WITH 3MM INC 625 CLAD FACE & BORE	0	0	
1560	U04		FLANGE	5557040_C60	10	24	FLG WN 600# RF LTCS A350 LF2 CL.1 SCH 80 WITH 3MM INC 625 CLAD FACE & BORE	0	0	
1570	U04	816	FLANGE	5557036_C60	4	8	FLG WN 600# RF LTCS A350 LF2 CL.1 SCH XS WITH 3MM INC 625 CLAD FACE & BORE	0	0	
1580	U04	817	FLANGE	5517251_C80	6	6	FLG WN 10000 API 6A TYPE 6BX 5 1/8" NS (6"NB) RTJ BORE 131MM LTCS A350 LF2 CL.1 SCH XS WITH 3MM INC 825 CLAD FACE & BORE	0	0	
1590	U04	817	FLANGE	5517251_C60	6	6	FLG WN 10000 API 6A TYPE 6BX 5 1/8" NS (6"NB) RTJ BORE 131MM LTCS A350 LF2 CL.1 WITH 3MM INC 625 CLAD FACE & BORE	0	0	
1600										
1600							***** ORIFICE FLANGES *****			
1610										
1610			ORIFICE FLG	5650840	2	3	WN ORIF SET 600# RF ALLOY 825 B564 UNS N08825 SCH 40S	0	0	
1620	U04		ORIFICE FLG	5624352_C80	4	8	WN ORIF SET 600# RF LTCS A350 LF2 CL.1 SCH XS WITH 3MM INC 825 CLAD FACE & BORE	0	0	
1630	U04		ORIFICE FLG	5624353_C80	10	16	WN ORIF SET 600# RF LTCS A350 LF2 CL.1 SCH 80 WITH 3MM INC 825 CLAD FACE & BORE	0	0	
1640	U04		ORIFICE FLG	5624348_C60	4	4	WN ORIF SET 600# RF LTCS A350 LF2 CL.1 SCH 120 WITH 3MM INC 625 CLAD FACE & BORE	0	0	
1650	U04		ORIFICE FLG	5624353_C60	10	16	WN ORIF SET 600# RF LTCS A350 LF2 CL.1 SCH 80 WITH 3MM INC 625 CLAD FACE & BORE	0	0	
1660	U04		ORIFICE FLG	5624352_C60	4	8	WN ORIF SET 600# RF LTCS A350 LF2 CL.1 SCH XS WITH 3MM INC 625 CLAD FACE & BORE	0	0	
1670										
1670		810					***** PIPE NIPPLES *****			
1680										
1680				0			NIPPLE PBE TO BE CUT FROM PIPE			
1690										
1690							***** GASKETS *****			
1700										
1700	U04		GASKET	5673661	0.5	24	GASKET 600# ALLOY 825 SPIRAL WOUND GRAPHITE IR	0	0	
1710	U04	817	GASKET	5673409	30	30	GASKET 600# RJ INC 718 ENERGIZED SEALRING COMPACT FLANGE LINE CLASS 600C80	0	0	
1720	U04		GASKET	5673800	2	2	GASKET 1500# ALLOY 825 SPIRAL WOUND GRAPHITE IR	0	0	
1730			GASKET	5674469	6	6	GASKET 10000 API 6A BX 169 5 1/8" NS ALLOY 625 RING	0	0	
1740										
1740		B1					***** BOLTING *****			
1750										
1750			BOLTING	5675770	0.5	99	STUD-BOLT A320 GR L7M/A194 GR 7M (S3) WITH 2 NUTS	0	0	
1760										
1760		SB6	BOLTING	5675675	10	16	JACK SCREW HEX HEAD A320 GR L7M	0	0	
1770										

	TENGIZCHEVROI TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES - BUTT WELD PWHIT FOR SIZES 4" AND LARGER SEAL WELDING - FOR THREADED CONNECTIONS ONLY PIPE BENDS - NONE THREAD SEALANT - PTFE PASTE	CORR. ALLOW. 1 MM	SOUR SERVICE - NACE MR0175/ ISO 15156 DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.3 FLUID CODES - PIM-SU-5112-TCO	MATERIAL: LTCS + CRA FLANGE RATING: 600 RF TEMPERATURE RANGE: -40 / 200 °C	12/16/2024	BY OZAC	U04	Page 3 of 4	600C80

SEQ NO	REV	NOTE NO.	ITEM NAME	ITEM CODE	SIZE RANGE / SMALL END SIZES		DESCRIPTION	REFERENCE / LARGE END SIZES		NOTES
1770		SB5					***** BLINDS & SPACERS *****			
1780										
1780			SPEC BLIND	5750230	0.75	3	SPECTACLE BLIND 600# RF ALLOY 825 B424 UNS N08825	0	0	
1790			SPEC BLIND	5749587_C80	4	8	SPECTACLE BLIND 600# RF LTCS A516-GR70 WITH 3MM INC 825 CLAD FACE & BORE	0	0	
1800	U04		SPEC BLIND	5749587_C60	4	8	SPECTACLE BLIND 600# RF LTCS A516-GR70 WITH 3MM INC 625 CLAD FACE & BORE	0	0	
1810										
1810			PADDLE	5750040_C80	10	16	PADDLE BLANK 600# RF LTCS A516-GR70 WITH 3MM INC 825 CLAD FACE	0	0	
1820	U04		PADDLE	5750040_C60	10	16	PADDLE BLANK 600# RF LTCS A516-GR70 WITH 3MM INC 625 CLAD FACE	0	0	
1830	U04		PADDLE	5750251_C60	10	16	PADDLE SPACER 600# RF LTCS A516-GR70 WITH 3MM INC 625 CLAD FACE & BORE	0	0	
1840										

	TENGIZCHEVROIL TCO TENGIZ, KAZAKHSTAN	CONSTRUCTION	CORR.ALLOW.	SERVICE	MATERIAL	DATE	BY	REV	SHEET	PIPING LINE CLASS
		ALL SIZES - BUTT WELD PWHT FOR SIZES 4" AND LARGER SEAL WELDING - FOR THREADED CONNECTIONS ONLY PIPE BENDS - NONE THREAD SEALANT - PTFE PASTE	CORR. ALLOW. 1 MM	SOUR SERVICE - NACE MR0175/ ISO 15156 DESIGN: ASME B16.5 MG 1.1 CODE: ASME B31.3 FLUID CODES - PIM-SU-5112-TCO	MATERIAL: LTCS + CRA FLANGE RATING: 600 RF TEMPERATURE RANGE: -40 / 200 °C	12/16/2024	BY OZAC	U04	Page 4 of 4	600C80